Today: one culture plate -> one assay plate

96 harvested wells -> 192 paired measurements + 30 spikes + 30 backgrounds + 42 standards = 294 assay wells

1 Measure growth

Mix consistently and read OD600 BEFORE clearing. Keep the six incubated medium controls and save raw well IDs.

2 Harvest separately

Clear using the established procedure. Keep all 96 supernatants separate. Do not pool the six culture replicates.

3 Prepare the pairs

Dilute in FRESH CRUDE-GLYCEROL SCREEN MEDIUM. Each assay well gets 5 uL supernatant + 10 uL medium.

4 Dispense 15 uL twice

First into the same coordinate; second one row BELOW, same column. Then add the extra checks and standards.

PRIMARY DUPLICATES / columns 2-17 only

Culture / stage	Assay 1	Assay 2
B	B	C
D	D	E
F	F	G
H	H	I
J	J	K
L	L	M

Example: culture B2 -> assay B2 AND C2

Each destination receives 15 uL of the SAME prepared dilution. The row shift is at the destination; keep the source fixed.

Six culture wells per condition x two aliquots each = six cultures measured twice. Compare each adjacent-row pair first.

Load only intended sample positions. Confirm tip footprint/clearance for the +1-row shift; keep extra controls out of that transfer path.

15 uL sample + 15 uL assay mix -> 1 h, room temperature, DARK

Then +15 uL 1 M acetic acid -> 45 uL final -> read A490 promptly

PURPLE BACKGROUND WELLS: use 15 uL plain 0.2 M Tris pH 8.2 in place of the assay mix. Add the same acid stop. Do not blanket-dispense assay mix over the whole plate.

The assay plate: exact destination wells

A at top; 1 at left. Uncoloured wells are unused. All measurements are on the same 384-well assay plate.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
A																								
B		M02 R1 / 1	M03 R1 / 1	M12 R1 / 1	M01 R1 / 1	M04 R1 / 1	C02 R1 / 1	M06 R1 / 1	C03 R1 / 1	M07 R1 / 1	M11 R1 / 1	C01 R1 / 1	M10 R1 / 1	M09 R1 / 1	M08 R1 / 1	BLK R1 / 1	M05 R1 / 1	M02 R1 / 1	M03 R1 / 1	M12 R1 / 1	M01 R1 / 1	M04 R1 / 1	W 3 std 3	
C		M02 R1 / 2	M03 R1 / 2	M12 R1 / 2	M01 R1 / 2	M04 R1 / 2	C02 R1 / 2	M06 R1 / 2	C03 R1 / 2	M07 R1 / 2	M11 R1 / 2	C01 R1 / 2	M10 R1 / 2	M09 R1 / 2	M08 R1 / 2	BLK R1 / 2	M05 R1 / 2	M02 R1 / 2	M03 R1 / 2	M12 R1 / 2	M01 R1 / 2	M04 R1 / 2	W 0.375 std 2	
D		M08 R2 / 1	BLK R2 / 1	M05 R2 / 1	M02 R2 / 1	M03 R2 / 1	M12 R2 / 1	M01 R2 / 1	M04 R2 / 1	C02 R2 / 1	M06 R2 / 1	C03 R2 / 1	M07 R2 / 1	M11 R2 / 1	C01 R2 / 1	M10 R2 / 1	M09 R2 / 1	C02 R1 / 1	M06 R1 / 1	C03 R1 / 1	M07 R1 / 1	M11 R1 / 1	W 1.5 std 3	
E		M08 R2 / 2	BLK R2 / 2	M05 R2 / 2	M02 R2 / 2	M03 R2 / 2	M12 R2 / 2	M01 R2 / 2	M04 R2 / 2	C02 R2 / 2	M06 R2 / 2	C03 R2 / 2	M07 R2 / 2	M11 R2 / 2	C01 R2 / 2	M10 R2 / 2	M09 R2 / 2	C02 R1 / 2	M06 R1 / 2	C03 R1 / 2	M07 R1 / 2	M11 R1 / 2	W 0.75 std 2	
F		C01 R3 / 1	M10 R3 / 1	M09 R3 / 1	M08 R3 / 1	BLK R3 / 1	M05 R3 / 1	M02 R3 / 1	M03 R3 / 1	M12 R3 / 1	M01 R3 / 1	M04 R3 / 1	C02 R3 / 1	M06 R3 / 1	C03 R3 / 1	M07 R3 / 1	M11 R3 / 1	C01 R1 / 1	M10 R1 / 1	M09 R1 / 1	M08 R1 / 1	M05 R1 / 1	M 12 std 2	
G		C01 R3 / 2	M10 R3 / 2	M09 R3 / 2	M08 R3 / 2	BLK R3 / 2	M05 R3 / 2	M02 R3 / 2	M03 R3 / 2	M12 R3 / 2	M01 R3 / 2	M04 R3 / 2	C02 R3 / 2	M06 R3 / 2	C03 R3 / 2	M07 R3 / 2	M11 R3 / 2	C01 R1 / 2	M10 R1 / 2	M09 R1 / 2	M08 R1 / 2	M05 R1 / 2	M 0 std 3	
H		C03 R4 / 1	M07 R4 / 1	M11 R4 / 1	C01 R4 / 1	M10 R4 / 1	M09 R4 / 1	M08 R4 / 1	BLK R4 / 1	M05 R4 / 1	M02 R4 / 1	M03 R4 / 1	M12 R4 / 1	M01 R4 / 1	M04 R4 / 1	C02 R4 / 1	M06 R4 / 1	M 0 std 2	W 0.375 std 3		W 3 std 1			
I		C03 R4 / 2	M07 R4 / 2	M11 R4 / 2	C01 R4 / 2	M10 R4 / 2	M09 R4 / 2	M08 R4 / 2	BLK R4 / 2	M05 R4 / 2	M02 R4 / 2	M03 R4 / 2	M12 R4 / 2	M01 R4 / 2	M04 R4 / 2	C02 R4 / 2	M06 R4 / 2	M 0.375 std 3	M 3 std 1		M 6 std 2	M 0.75 std 3	W 0 std 3	
J		M04 R5 / 1	C02 R5 / 1	M06 R5 / 1	C03 R5 / 1	M07 R5 / 1	M11 R5 / 1	C01 R5 / 1	M10 R5 / 1	M09 R5 / 1	M08 R5 / 1	BLK R5 / 1	M05 R5 / 1	M02 R5 / 1	M03 R5 / 1	M12 R5 / 1	M01 R5 / 1		W 3 std 2	M 0.75 std 2	W 6 std 2	M 0.375 std 1		
K		M04 R5 / 2	C02 R5 / 2	M06 R5 / 2	C03 R5 / 2	M07 R5 / 2	M11 R5 / 2	C01 R5 / 2	M10 R5 / 2	M09 R5 / 2	M08 R5 / 2	BLK R5 / 2	M05 R5 / 2	M02 R5 / 2	M03 R5 / 2	M12 R5 / 2	M01 R5 / 2		M 3 std 3	W 6 std 3	M 1.5 std 2	W 12 std 3	M 6 std 3	
L		M03 R6 / 1	M12 R6 / 1	M01 R6 / 1	M04 R6 / 1	C02 R6 / 1	M06 R6 / 1	C03 R6 / 1	M07 R6 / 1	M11 R6 / 1	C01 R6 / 1	M10 R6 / 1	M09 R6 / 1	M08 R6 / 1	BLK R6 / 1	M05 R6 / 1	M02 R6 / 1		M 1.5 std 1		W 6 std 1		W 0 std 2	
M		M03 R6 / 2	M12 R6 / 2	M01 R6 / 2	M04 R6 / 2	C02 R6 / 2	M06 R6 / 2	C03 R6 / 2	M07 R6 / 2	M11 R6 / 2	C01 R6 / 2	M10 R6 / 2	M09 R6 / 2	M08 R6 / 2	BLK R6 / 2	M05 R6 / 2	M02 R6 / 2		M 12 std 1	W 0.75 std 1	M 1.5 std 3	M 0 std 1	M 6 std 1	
N		M02 R1 / 1	M03 R1 / 1	M12 R1 / 1	M01 R1 / 1	M04 R1 / 1	C02 R1 / 1	M06 R1 / 1	C03 R1 / 1	M07 R1 / 1	M11 R1 / 1	C01 R1 / 1	M10 R1 / 1	M09 R1 / 1	M08 R1 / 1	M05 R1 / 1	W 0 std 1	W 12 std 2	M 3 std 2		W 0.375 std 1	W 1.5 std 2	M 0.75 std 1	
O		M02 R1 / 2	M03 R1 / 2	M12 R1 / 2	M01 R1 / 2	M04 R1 / 2	C02 R1 / 2	M06 R1 / 2	C03 R1 / 2	M07 R1 / 2	M11 R1 / 2	C01 R1 / 2	M10 R1 / 2	M09 R1 / 2	M08 R1 / 2	M05 R1 / 2			M 0.375 std 2	W 12 std 1	M 12 std 3	W 1.5 std 1	W 0.75 std 3	
P																								

Unspiked +3 mM spike Background: Tris only Standard: W water / M medium

M01 256 | M02 75 | M03 269 | M04 294 | M05 275 | M06 41 | M07 158 | M08 145 | M09 200 | M10 196 | M11 291 | M12 264

C01: 41+200+196 C02: 256+269+294+41+145+200 C03: 294+275 BLK: incubated medium

R = culture replicate; /1 and /2 = analytical aliquots. Standard concentrations are mM BEFORE assay reagents.

Spikes + backgrounds: use culture replicate 1

Take these checks from the original row-B cultures. B16 is the incubated medium control and has primary duplicates only.

Prepare this volume	Supernatant	12 mM medium spike	Plain fresh medium	Dispense
Ordinary sample pair	10 uL	0	20 uL	15 + 15 uL
R1: unspiked + background	20 uL	0	40 uL	4 x 15 uL
R1: spike pair	10 uL	7.5 uL	12.5 uL	15 + 15 uL

These are minimum recovered volumes, before handling losses. R1 checks require 30 uL original supernatant per selected culture; other wells require 10 uL. Prepare extra at the same ratios for your actual staging/dead volume.

Culture	Condition / members	Unspiked pair	Spike pair	Background pair
B2	M02 (75)	B2 / C2	N2 / O2	B18 / C18
B3	M03 (269)	B3 / C3	N3 / O3	B19 / C19
B4	M12 (264)	B4 / C4	N4 / O4	B20 / C20
B5	M01 (256)	B5 / C5	N5 / O5	B21 / C21
B6	M04 (294)	B6 / C6	N6 / O6	B22 / C22
B7	C02 (256+269+294+41+145+200)	B7 / C7	N7 / O7	D18 / E18
B8	M06 (41)	B8 / C8	N8 / O8	D19 / E19
B9	C03 (294+275)	B9 / C9	N9 / O9	D20 / E20
B10	M07 (158)	B10 / C10	N10 / O10	D21 / E21
B11	M11 (291)	B11 / C11	N11 / O11	D22 / E22
B12	C01 (41+200+196)	B12 / C12	N12 / O12	F18 / G18
B13	M10 (196)	B13 / C13	N13 / O13	F19 / G19
B14	M09 (200)	B14 / C14	N14 / O14	F20 / G20
B15	M08 (145)	B15 / C15	N15 / O15	F21 / G21
B17	M05 (275)	B17 / C17	N16 / O16	F22 / G22

SPIKE: +3 mM added lactate in the 15 uL aliquot (+1 mM after stop). Use 12 mM lactate in the SAME fresh medium; replace medium volume with spike, do not add it on top. Mix each preparation before splitting.

Standards, reagents and reading

Use the existing high-LDH formulation. Add standards and supplemental checks after the primary offset transfers.

mM*	Fresh-medium standards	Water standards
0	M22 / H18 / G23	N17 / L23 / I23
0.375	J22 / O19 / I18	N21 / C23 / H19
0.75	N23 / J20 / I22	M20 / E23 / O23
1.5	L19 / K21 / M21	O22 / N22 / D23
3	I19 / N19 / K19	H21 / J19 / B23
6	M23 / I21 / K23	L21 / J21 / K20
12	M19 / F23 / O21	O20 / N18 / K22

*Before assay reagents. Dispense 15 uL per listed well; 45 uL per level/matrix. All get 15 uL assay mix + 15 uL acid stop.

Serial standards: prepare each matrix separately

Start with 200 uL of 12 mM. Put 100 uL matching blank into each of five tubes. Transfer 100 uL sequentially, mixing each time: 12 -> 6 -> 3 -> 1.5 -> 0.75 -> 0.375 mM. Keep a separate 0 mM blank.

Reserve 112.5 uL of the 12 mM MEDIUM stock for all 30 spike wells, plus handling losses. Do not use up the spike stock in the serial dilution.

For 1 mL of 12 mM stock: 12 uL of 1 M lactate in 1,000 uL FINAL volume. For medium stock, count stock water within the medium recipe so medium remains 1x. Use water for the separate water series.

ASSAY MIX: existing ~6.4 mL SOP batch

0.2 M Tris pH 8.2	**5,440 uL**
NAD+ stock, 22 mg/mL	**640 uL**
INT stock, 10 mg/mL	**320 uL**
LDH stock, 2.75 U/uL	**7.68 uL**
Methoxy-PMS stock, 25 mg/mL	**1.41 uL**

Freshly prepare and protect from light. Stocks and solvents follow the local assay SOP. Total dispensed mix: 3.960 mL (264 wells).

Dispensed totals (exclude reservoir dead volume)

Assay mix: 3,960 uL | Plain Tris: 450 uL

1 M acetic acid stop: 4,410 uL (all 294 wells)

Check supply volume against the dispensing method. The 6.4 mL batch does not guarantee enough liquid for every reservoir/head setup.

READ A490 promptly after stop; save the raw export and record OD, harvest, assay-start, stop and read times. Use the same reader method across the plate.

Compare paired measurements first, then spike recovery, then variation between the six culture wells.